

Augmenting Karpathy's LLM Council with Formal Decision Evaluation Methods

AI Ideation Run

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Ideas: 1

Depth: Comprehensive

The Structured Council — A Decision Architecture That Combines Multi-Model Diversity with Multi-Framework Rigor

Merge the LLM Council's model-diversity approach with the Decision Evaluation Framework's analytical-diversity approach, creating a system where multiple LLMs each apply multiple formal decision frameworks, producing a two-dimensional deliberation matrix that surfaces insights neither system can generate alone.

The Core Problem Both Systems Partially Solve

Both Karpathy's LLM Council and the Decision Evaluation Framework attack the same fundamental problem from different angles: single-perspective analysis produces blind spots. The Council solves this by querying multiple models — different training data, different RLHF tuning, different reasoning patterns. The Decision Framework solves it by applying multiple analytical lenses — economic, strategic, temporal, probabilistic, values-based. But each system only diversifies along one axis.

The Council asks the same question to different models. It gets perspective diversity but not analytical diversity — all four models might focus on the same aspects of the problem, just with different conclusions. The Decision Framework applies different frameworks to the same question via the same model. It gets analytical diversity but not perspective diversity — all seven frameworks are interpreted through a single model's training biases.

The synthesis opportunity is a **two-dimensional deliberation matrix**: M models $\times$ F frameworks, producing $M \times F$ independent analyses before any synthesis occurs. With 4 models and 7 frameworks, that's 28 independent analytical cells — each one a unique combination of “who is thinking” and “how they're thinking.”

Architecture: Four Stages Instead of Three

The current LLM Council runs three stages: (1) independent responses, (2) anonymous peer review, (3) chairman synthesis. The augmented system would restructure this into four stages that interleave the two approaches:

Stage 1: Framework-Structured Independent Analysis

Instead of asking each council model an open-ended question, present the decision alongside all seven framework methodologies. Each model independently produces seven framework analyses. This replaces the Council's unstructured Stage 1 with structured analysis that ensures every model covers every analytical dimension.

The key insight is that different models will produce different outputs even when given the same framework. Ask GPT, Claude, Gemini, and Grok to each run a Cost-Benefit Analysis on the same decision, and you'll get four different cost-benefit analyses — because each model has different assumptions about what counts as a cost, how to estimate magnitudes, and which indirect effects matter. The framework provides structure; the model provides judgment.

This stage produces a 4×7 matrix of independent analyses, each with a 0-100 score and supporting rationale.

Stage 2: Cross-Model Framework Comparison

For each of the seven frameworks, compare results across all four models. This is the equivalent of the Council's peer review stage, but structured by framework. For each framework, the system identifies:

- **Consensus scores:** Where all models produced similar scores (e.g., all four rate Cost-Benefit between 70-80), the framework's signal is robust.
- **Divergent scores:** Where models disagree significantly (e.g., Risk-Reward scores of 30, 45, 72, 68), the framework is revealing genuine uncertainty or assumption sensitivity.
- **Assumption divergence:** The specific assumptions driving score differences. If Model A rates financial costs much higher than Model B for Cost-Benefit, that assumption difference is surfaced explicitly.

This stage also runs the Council's anonymous peer review: each model reviews the other models' framework analyses (anonymized) and ranks them for rigor, completeness, and insight quality. This catches cases where a model's framework analysis is technically correct but superficial.

Stage 3: Cross-Framework Pattern Analysis

Now flip the axis. For each model, compare its seven framework scores against each other. This reveals the internal tensions that the Decision Framework is designed to surface:

- **Framework agreement:** If a model's Cost-Benefit, Decision Matrix, and ICE scores all point the same direction, the economic case is clear.
- **Framework conflict:** If Cost-Benefit says "strong yes" but Regret Minimization says "strong no," the decision has a values-versus-economics tension.
- **Temporal patterns:** If Eisenhower says "urgent" but Regret Minimization shows low long-term regret for inaction, the urgency may be artificial.

The critical addition is comparing these patterns across models. If all four models show the same framework conflict (e.g., all show Cost-Benefit vs. Regret Minimization tension), that tension is likely inherent to the decision. If only one model shows it, it may be an artifact of that model's training biases.

Stage 4: Structured Chairman Synthesis

The Chairman model (or a designated synthesis model) receives:

1. The full 4×7 matrix of scores
2. Stage 2's cross-model consensus and divergence maps per framework
3. Stage 3's cross-framework tension patterns per model and across models
4. All peer review rankings from Stage 2

The synthesis output is structured as:

- **Decision Landscape:** What the 28 analyses collectively reveal about the decision's character. Is this primarily an economic decision? A values decision? A timing decision?
- **Areas of Robust Consensus:** Where both model diversity and framework diversity converge — the aspects of the decision that are genuinely clear.
- **Areas of Genuine Uncertainty:** Where either models disagree within a framework or frameworks disagree within a model — the aspects requiring human judgment.
- **Hidden Trade-offs:** Tensions revealed by cross-framework analysis that weren't apparent in the original question framing.
- **Recommended Investigation Points:** Specific assumptions or factors that, if resolved, would most reduce the remaining uncertainty.

The Chairman explicitly does NOT produce a single recommendation. Following the Decision Framework's philosophy, it illuminates the decision landscape and identifies where human judgment is most needed.

Why This Combination Is More Than the Sum of Its Parts

The individual systems have specific blind spots that the combination addresses:

Blind spot 1: The Council’s analytical shallowness. When asked an open-ended question, LLMs tend to focus on the most salient aspects and may all gravitate toward similar framings. Forcing each model through seven frameworks ensures comprehensive coverage. A model that naturally focuses on financial aspects is forced to also consider temporal urgency (Eisenhower) and long-term regret (Regret Minimization).

Blind spot 2: The Decision Framework’s single-model bias. When one model runs all seven frameworks, its biases pervade every analysis. If Claude systematically underweights competitive dynamics, its SWOT analysis, Decision Matrix, and Risk-Reward assessment will all share that blind spot. Running each framework across four models means that bias gets diluted and, more importantly, becomes detectable through cross-model comparison.

Blind spot 3: Both systems’ synthesis limitations. The Council’s Chairman synthesis is unstructured — it can miss important patterns in the Stage 1 responses. The Decision Framework doesn’t synthesize at all — it presents seven scores and leaves interpretation to the user. The augmented system’s four-stage approach structures synthesis through intermediate comparison steps (Stages 2 and 3) before final synthesis (Stage 4), making the Chairman’s job tractable and the output interpretable.

Blind spot 4: Distinguishing model uncertainty from decision uncertainty. When the current Council’s models disagree, it’s ambiguous whether they disagree because the question is genuinely hard or because their training differs. The framework-structured approach partially resolves this: if models disagree on Cost-Benefit scores but agree on SWOT and Risk-Reward, the disagreement is likely about specific cost/benefit estimates (model uncertainty) rather than about the decision’s overall character (decision uncertainty).

Implementation Approach

The technical integration is feasible because both systems share key architectural patterns:

- **Parallel execution:** The Council queries models in parallel via async httpx; the Decision Framework runs frameworks in parallel via Claude Code Tasks. The combined system would nest these: parallel models, each running parallel frameworks.
- **Independence before synthesis:** Both systems explicitly keep analyses independent until a dedicated synthesis step. The combined system extends this principle with two intermediate comparison steps before final synthesis.
- **Structured output:** Both systems produce structured, scored output rather than free-form text. The combined system’s 4×7 matrix is a natural extension.

The primary implementation would extend the Council’s FastAPI backend with:

1. A framework definitions endpoint serving the seven methodology guides
2. Modified Stage 1 prompts that include framework instructions
3. New Stage 2 and Stage 3 comparison logic (largely pattern matching on the score matrix)
4. Enhanced Stage 4 Chairman prompt with the structured intermediate outputs

The frontend would add a matrix visualization showing the 4×7 grid with color-coded scores, highlighting consensus cells (green), divergence cells (red), and the specific tensions identified in Stage 3.

Worked Example: Should a Startup Pivot to Enterprise Sales?

To illustrate the difference, consider a startup deciding whether to pivot from self-serve to enterprise sales:

Council alone might produce: GPT emphasizes revenue potential, Claude emphasizes execution risk, Gemini emphasizes market timing, Grok emphasizes competitive dynamics. The Chairman synthesizes these into a balanced perspective. But all four models might miss the founder regret dimension or the opportunity cost framing.

Decision Framework alone might produce: Cost-Benefit shows strong unit economics for enterprise; SWOT reveals weak enterprise sales team; Decision Matrix scores enterprise high on revenue but low on team fit; ICE shows high impact but low ease; Risk-Reward shows asymmetric upside; Eisenhower says important but not urgent; Regret Minimization shows high regret for inaction at the 5-year mark. Rich analysis, but filtered through one model's assumptions about enterprise sales economics.

Combined system produces: The 4×7 matrix reveals that all four models agree on strong Cost-Benefit (consensus: enterprise unit economics are favorable) but disagree sharply on Risk-Reward (GPT: 78, Claude: 42, Gemini: 65, Grok: 55). Investigation shows the disagreement stems from different assumptions about enterprise sales cycle length. Meanwhile, cross-framework analysis shows that all models exhibit tension between ICE (low ease) and Regret Minimization (high inaction regret) — revealing that the core tension isn't whether to pivot but how to sequence the transition to manage execution risk. This insight — that the real question is “how” not “whether” — might not emerge from either system alone.

Extensions and Variations

Weighted frameworks by decision type: Not all seven frameworks are equally relevant to every decision. For financial decisions, upweight Cost-Benefit and Risk-Reward. For career decisions, upweight Regret Minimization and Eisenhower. The system could auto-detect decision type and adjust framework weights in the synthesis.

Adversarial framework assignment: Instead of all models running all frameworks, assign each model a subset of frameworks and have them critique each other's framework analyses. This reduces computational cost (7 analyses per model instead of 7) while maintaining the cross-model diversity benefit.

Temporal re-evaluation: Run the combined analysis at decision time, then re-run it 3, 6, and 12 months later with updated information. Track how the 4×7 matrix evolves over time to calibrate both the models and the frameworks.

Framework evolution: Use the cross-model comparison data to identify frameworks that consistently produce the most divergent scores across models. These are either the most informative frameworks (because they surface genuine uncertainty) or the most poorly specified (because models interpret them differently). Either way, they're the ones most worth refining.

Domain-specific framework substitution: Replace general frameworks with domain-specific ones for specialized decisions. For investment decisions, swap Eisenhower Matrix for a Discounted Cash Flow framework. For hiring decisions, swap Cost-Benefit for a Culture Fit / Culture Add framework. The architecture supports this because frameworks are defined as methodology guides, not hard-coded logic.

Alternatives

Alternative 1: Sequential rather than parallel. Instead of the 4×7 matrix, run the Decision Framework first (single model, 7 frameworks), then submit the framework outputs to the Council for multi-model critique. Simpler but loses the benefit of independent parallel analysis — later models are anchored by the first model’s framework outputs.

Alternative 2: Council-of-frameworks without multi-model. Run all seven frameworks on a single model but treat each framework as a “council member” with its own vote. Peer review becomes cross-framework critique (does the Cost-Benefit analysis account for the risks identified in Risk-Reward?). Cheaper and simpler but loses model diversity entirely.

Alternative 3: Ensemble scoring. Instead of the structured four-stage approach, simply average the 4×7 matrix into seven framework scores and one overall score. Computationally equivalent but throws away the most valuable information — the disagreement patterns that reveal where human judgment is needed.

Alternative 4: Human-in-the-loop at Stage 2. Instead of automated cross-model comparison, present the raw 4×7 matrix to a human decision-maker after Stage 1 and let them direct the investigation. Higher quality synthesis but much slower and defeats the purpose of automated analysis for routine decisions.

Category: AI Infrastructure / Decision Science

Feasibility: Medium — Both component systems exist and work; the integration requires non-trivial orchestration and prompt engineering but no novel AI capabilities. The main challenge is managing the computational cost (28 LLM calls per decision at Stage 1 alone) and designing Stage 2-3 comparison logic that reliably extracts meaningful patterns from the score matrix. Estimated 4-6 weeks of engineering effort to build a working prototype on top of the existing Council codebase.

Impact: High — Decision quality improvement compounds over time. For businesses making dozens of strategic decisions per quarter, even marginal improvement in decision quality has outsized returns. For individuals facing life-defining choices (career pivots, major investments, relocations), the structured illumination of trade-offs and uncertainties directly addresses the most common sources of decision regret: not considering enough perspectives and not recognizing hidden tensions until after committing.

Key Considerations:

- Computational cost scales as $M \times F$ per decision — 28 calls for 4 models × 7 frameworks, plus peer review calls. Need to consider whether all decisions warrant full analysis or whether a tiered approach (quick: 2 models × 3 frameworks; standard: 4×7; deep: 6×7 with adversarial review) is more practical.
- The quality of framework-structured prompts is critical. Poorly specified framework instructions will produce generic analyses regardless of model diversity. The Decision Framework’s existing methodology guides provide a strong starting point.
- Score calibration across models is non-trivial. Model A’s “75” may not mean the same thing as Model B’s “75.” The cross-model comparison in Stage 2 partially addresses this by looking at relative patterns rather than absolute scores.
- The system’s output is complex — a 4×7 matrix with multiple layers of comparison. The synthesis in Stage 4 must compress this into actionable insight without losing the nuance. Frontend visualization (heatmaps, tension diagrams) is essential for usability.

- Risk of analysis paralysis: presenting 28 independent analyses plus meta-analyses could overwhelm rather than inform. The Stage 4 synthesis must aggressively prioritize the 2-3 most important insights.

Next Steps:

1. Fork the LLM Council repository and add the Decision Framework's seven methodology guides as framework definition files.
2. Modify the Stage 1 prompt template to include framework instructions, producing structured 0-100 scored outputs per framework per model.
3. Build Stage 2 and Stage 3 comparison logic as Python modules that consume the 4×7 matrix and output structured consensus/divergence reports.
4. Design the Stage 4 Chairman prompt with explicit instructions for structured synthesis (Decision Landscape, Consensus, Uncertainty, Trade-offs, Investigation Points).
5. Build a matrix visualization component in the React frontend showing the 4×7 grid with interactive drill-down into individual cells.
6. Test on 10 well-documented historical decisions where outcomes are known, validating that the combined system surfaces insights that neither component system produces alone.

Generated by Claude (Anthropic) as part of a structured AI ideation run.